

The Trawl — Arcade Game 2 Design Document (Draft)

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Overview and pillars

The Trawl is a 1-6 player co-op fishing horror game: a crew works a steam trawler through three nights per deadline to meet the Owners' quota, while a living food web under the hull decides what bites, what hunts, and what comes up the rail. It replaces the Trawl section of the Depth Master Reference (1-4 players, one stub loop) and becomes the full design Claude Code builds from, the way Fathoms has its own document.

What it takes from each source. From Lethal Company: the quota and deadline, a shared ship, scavenging for value, a store and a buy-back rate that changes by day, monsters with tells and counters, a terminal, proximity voice, and dead players who stay useful. From How to Fish: a real cast-and-reel with line tension, rod and reel stats, lures and bait that matter per species, fishing grounds with their own catch tables, a collection log, and gear progression that opens new water.

Pillars

1. **The sea is simulated, not scripted.** Fish, predators, and threats are agents in one food web. A monster shows up because something drew it: blood in the water, light, noise, a school it hunts.
2. **Every catch is a physics fight.** Line, rod, fish, and boat are one connected system. A marlin can drag the boat; a snagged net can heel it over.
3. **Greed is the enemy.** Brighter lanterns, bigger nets, chum, and deeper water all pay more and all attract what kills you. The crew decides every night how far to push.
4. **Six people, one boat, too many jobs.** There are always more stations than hands, so talking, trading jobs, and trusting each other is the game.
5. **Dread is readable.** Every threat has a sound tell, a visual tell, a sonar tell, and a counter. Death is always explainable afterward.

Scope rules. Everything stays procedural (no image or audio files). The deck is top-down pixel art from the parkour pipeline; underwater views (sonar, dive, reel camera) use the same pixel palette. Numbers live in `trawl_data.cpp`. The host runs the simulation; clients send commands. Every system ships with a headless self-test on `depth.exe`.

The session loop

A run is a string of deadlines. Each deadline is three nights at sea; a session (one deadline) takes about 30–35 minutes with six players. Meet the quota and the next deadline starts with a higher one; miss it and the Owners repossess the boat and the run ends.

The Owners. An unseen shipping concern that rents the crew the *Gannet*, a 22 m steam trawler, and licenses it to fish only at night. They speak through a brass telegraph in the wheelhouse (the Lethal Company terminal and Company voice in one). Their messages are terse, occasionally wrong, and never sympathetic.

The telegraph. The Owners speak only by brass telegraph: a tape clatters out of the wheelhouse printer in capitals, with STOP for every full stop, and a bell tick every crewman hears. Their voice is mercantile, polite in form, indifferent in fact; they care about the hold and the boat, never the hands. Everything the Owners say is printed on tape that can be torn off and read later on the crew board. About 30% of rumors are wrong, and the Owners never admit it.

When	Example message
A deadline begins	QUOTA 400 SHILLINGS STOP THREE NIGHTS STOP THE OWNERS ARE CONFIDENT STOP
A night begins	WEEDS FOG STOP MOON WANING STOP WHALERS REPORT TUNA ON SEAWARD EDGE STOP
The Grotto closing time	ARCH IMPASSABLE AFTER 0315 STOP OWNERS ACCEPT NO RESPONSIBILITY STOP
Midnight	MIDNIGHT STOP HOLD INSPECTED AT HARBOUR LINE 0500 STOP
04:00	ONE HOUR STOP CUSTOMS CUTTER ON STATION STOP
A fine	LANDING OF PROTECTED TURTLE NOTED STOP 50 SHILLINGS DEDUCTED STOP
Hull damage below 40 in any section	KINDLY REFRAIN FROM DAMAGING OWNERS PROPERTY STOP
A first catch of a species	SPECIMEN NOTED STOP BONUS AUTHORISED STOP DO NOT BRUISE STOP

When	Example message
A crewman dies	HAND DECEASED STOP REPLACEMENT CHARGED IF REMAINS NOT RECOVERED STOP
Deadline met	QUOTA MET STOP NEW QUOTA 620 STOP THE OWNERS EXPECTED NOTHING LESS STOP
Deadline missed	QUOTA NOT MET STOP GANNET REPOSSESSED STOP CREW RELEASED WITHOUT REFERENCE STOP

The telegraph as a tell. On the Kraken's Glass, the tape prints only dots. When the Ghost Ship is near, it prints a message from a boat that sank years ago. Neither is ever explained.

One deadline

1. **Dock (untimed, 2–4 min).** The chalkboard shows the quota, nights left, and money. The crew sells last night's catch, buys gear at the Chandler, refits the boat at the Slipway, and loads bait, ice, and coal.
2. **Choose a ground.** The helm picks one of the four grounds on the chart table. Each costs coal to reach, and the telegraph posts that night's conditions: weather, moon phase, and a rumor ("Whalers report the Weeds quiet"; rumors are right 70% of the time).
3. **Sail out (about 45 s).** A short top-down run through the harbor mouth and fog. The engine crew brings up steam; the helm avoids buoys and rocks. Anything left unsecured on deck slides.
4. **The night (9 real minutes, 20:00 to 05:00 on the wheelhouse clock).** The crew fishes, trawls, shoots, dives, and survives. The sea starts calm and wakes as the night goes on and as the crew makes blood, noise, and light.
5. **Run for harbor.** The crew decides when to leave. The return takes 45–90 s depending on distance and steam. The *Gannet* must cross the harbor line before 05:00; after that, the Owners' customs cutter seizes the hold and fines 10% of money. There is always a reason to stay ten more minutes.
6. **Sell.** Fish go to the Fish Market (paid now, price set by freshness and glut). Salvage goes to the Owners (paid at a rate that rises toward the deadline, see Economy). Money past the quota buys gear.
7. **Repeat** for nights two and three. After the third night's sale, the Owners count the quota.

Quota. Starts at 400 shillings for six players (scaled 0.5× solo, 0.7× for two, 0.85× for four; bots count as crew). After each met deadline it rises by 40% plus 60 shillings, so by the fourth deadline the crew must work Atlantis Waters or dive for relics to keep up. Surplus money carries over; the quota does not reset.

The ground remembers. A ground's ecosystem state (populations, hunger, blood, and any Wake the crew raised) persists across the three nights of a deadline. Emptying the Weeds of sardines on night one means hungrier, bolder predators there on night two. Grounds reset when a new deadline starts.

Night pacing (the Stir clock). Each ground has a baseline threat curve that rises through the night, plus whatever the crew adds. The first two minutes are almost always safe; the last two minutes of an ordinary night have a large threat on the water about half the time.

First night: the shakedown

The first time a lobby includes a player with no completed nights, the host is offered a shakedown night: a 6-minute night on Eclipse Lagoon with a fixed seed, no quota, and no deaths that count. It can be skipped, and any crew can replay it from the arcade menu. The only guide is Kess, an old deckhand who is aboard for this night only and talks in short chalked lines on the deck ("Line's slack. Reel.").

1. **The deck.** Walk to each station; take the handline; catch three silversides for the bait well.
2. **The rod.** Bait a light rod with a silverside, cast at the marked snapper school, set the hook on the take, and land a snapper with side pressure. Kess gaffs it if the angler asks with a ping.
3. **The table.** Gut and ice the snapper; Kess throws the guts over the side.
4. **The sonar.** Ping, mark the sardine ball, and steer onto it.
5. **The net.** Shoot the trawl through the ball and haul it with two hands; sort the moon jellies back over the side.
6. **Blood.** The guts from step 3 bring a reef shark. It takes a hooked fish, then circles. The crew learns the counters: stop the blood, get off the rail, gaff or cannon, or steam away.
7. **Overboard.** Kess slips on the wet deck and goes in. Stop the screw and throw the ring.
8. **Home.** The clock runs to 05:00 fast; cross the harbour line, carry the catch to the Fish Market, and read the chalkboard.

The Field Manual. A battered book in the wheelhouse holds a page per station, the tackle and bait basics, and the Catch Log with its threat dossiers. It fills in as the crew discovers things, like Lethal Company's terminal bestiary.

The crew of six

Each crewman takes a light role with a small perk, but any hand can still work any station, and there are twelve stations for six people, so the crew is always choosing what to leave unmanned. A station works only while someone stands at it; walking away from a reel mid-fight hands the fish the line.

Roles. Picked in the lobby; two players may take the same role, but its perk does not stack. The perks make a hand better at a job, never the only one who can do it.

Role	Perk	Starts with
Bosun	Winch, pump, and boiler shovel work 25% faster; patches leaks in 3 s instead of 6; sees hull section integrity on the HUD	Patch kit
Angler	Hook-set window +75 ms; sees the species name on the bite; line tension gauge shows the break point	Medium rod
Diver	+15 s of helmet air; no Bends on a normal-speed recall; moves 15% faster and carries salvage 25% faster in dives	Knife and helmet lamp
Medic	Bandages stop bleeding in 2 s instead of 6; can treat one serious injury per night (sets a broken arm, removes a hook); revives a drowned crewman with CPR if they are hauled aboard within 15 s of drowning	Medical bag (does not take an inventory slot)

Role upgrades. A role ranks up three times in a run: after the first, second, and fourth met deadlines. At each rank the player picks one of two upgrades on the dock chalkboard. Upgrades last until the run ends, survive death, and belong to the role: switching roles at the dock starts the new role at rank 1. Bots pick from a fixed preference list per role.

Role	Rank 2 (after deadline 1)	Rank 3 (after deadline 2)	Rank 4 (after deadline 4)
Bosun	Shipwright: a patched section regains 30 integrity · or Stoker: overpressure grace 5 → 10 s, screw one noise step quieter	Old Salt: never slides on deck; crew within 4 m slide 5° later · or Deck Boss: bots under their orders work 20% faster	Iron Hull: +25 max integrity on every section · or Full Steam: +20% speed for 20 s, once a night
Angler	Light Touch: slack tolerance 1.5 → 3 s · or Heavy Hand: side pressure tires fish 25% faster	Reader: the rod ticks 0.5 s before each run · or Strong Line: +20% line strength on any rod they hold	Trophy Hunter: first catches pay a further +50% · or Bait Master: their bait and lures keep scent twice as long and draw 25% more bites
Diver	Deep Lungs: +15 s more air · or Glint Eye: sees salvage through one wall	Pressure Hardened: hardhat works 30 m deeper; free-dives 20 s without a hose · or Wreck Rat: opens locked cabins without a crowbar	Strongback: carries two-diver lifts alone · or Old Hand: their hose and lifeline never tangle
Medic	Field Surgeon: treats two serious injuries a night · or Warm Blankets: overboard timer +8 s for the whole crew	Second Wind: CPR window 15 → 30 s · or Steady Nerves: charmed or stunned crew within 6 m recover twice as fast	Miracle Worker: once per deadline, revives a crewman whose body is aboard, mid-night · or Ghost Speaker: hears and talks with the dead, and sees ghosts' outlines

Upgrade numbers live in `trawl_data.cpp` with the rest; `--trawl-sim` reports quota rates with and without upgrades so no single pick dominates its pair.

Bots fill empty berths if the host chooses. In the lobby the host sets the crew size from 1 to 6 and whether empty berths get bots. A bot takes a role, works the station it is ordered to through the ping wheel ("Take the pump," "Take port rod," "Follow me"), and falls back to a sensible default watch when no one gives orders. Bots fish at a competent-but-not-expert level, never dive unless ordered, and do not use voice. A dropped player's seat is taken by a bot after the 2-minute reconnect window, per the shared arcade rule.

Bot crew

A bot is a crewman with a role, an inventory, and a body that can fall overboard, driven by a small utility brain in `trawl_bots.cpp`. It scores every station it could work each second, weighted by its role, standing orders, and what is happening, and walks to the best one. Bots are good hands, not good captains: they never decide where the boat goes or when to go home.

Skill. The host picks one of three levels for all bots.

Skill	Reaction time	Hook-set success	Gaff success	Aim error (cannon, rifle)
Green	700 ms	50%	60%	$\pm 4^\circ$
Able	400 ms	70%	80%	$\pm 2^\circ$
Old Hand	250 ms	85%	90%	$\pm 1^\circ$

Default watch by role (used when no one gives orders): Bosun bot on the boiler, moving to the pumps whenever there is water; Angler bot on a rod; Diver bot on the net winch and gutting table; Medic bot on the gutting table, watching the rails.

What a bot does at each station

Station	Bot behaviour
Helm	Holds heading and speed as ordered; holds station over a marked wreck or diver; never chooses a course on its own
Boiler	Keeps pressure in the green; bleeds at the first red
Pumps	Pumps until the section is dry
Rods	Casts the rigged lure at the nearest marked contact or a school it can see; fights by the model with side pressure; asks for a gaff with a ping
Net winch	Shoots and hauls only when ordered; cuts a snagged net only when ordered
Lantern	Leaves it where it is unless ordered

Station	Bot behaviour
Sonar	Pings every 6 s, marks the largest contact, marks any threat
Harpoon cannon	Fires only at threats, or at a hooked giant when ordered
Gutting table	Guts and ices landed fish, heaviest first; throws bycatch back alive
Air pump	Pumps in rhythm; never leaves while a diver is down

Reactions that override the watch

- **Overboard:** the nearest bot throws the life ring (with its aim error), the Medic bot runs to the rail, and the helm bot stops the screw.
- **The bell:** every bot moves to the nearest defensive post (cannon, gaff at the rail, rifle if carried) and looks for the threat.
- **Boarders:** bots with gaffs fight; others back away from the rail.
- **Leak or fire:** the Bosun bot, or the nearest bot if there is no Bosun, goes to it.
- **Charmed or grabbed crewman:** the nearest bot rings the bell, then helps.

Orders (ping wheel, aimed at a bot or at a station): take this station; follow me; hold here; haul the net; cut the line; fire on my mark; dive (Diver-role bots only); go home (all bots stop fishing and secure the deck).

What bots never do unless ordered: dive, use depth charges or explosive harpoons, board derelicts or the Ghost Ship, raise the lantern, or ring the bell for no reason. Bots do not use voice; they show short text barks ("Fish on, port!", "Water in the hold!") above their heads and in the ping feed. A dead bot's ghost rings the bell when a threat comes within 10 m.

The *Gannet's* stations

Station	Where	What it does	What happens if nobody's there
Helm	Wheelhouse	Steers, sets the engine telegraph (stop, slow, half, full)	The boat drifts with wind and current; a hooked marlin tows it

Station	Where	What it does	What happens if nobody's there
Boiler	Engine room, below	Shovel coal, bleed pressure, keep the gauge in the green	Pressure falls, speed drops; overpressure blows a valve (a fire and a 20 s stop)
Bilge pumps	Engine room	Pump water out after a breach	The boat settles, lists, and sinks if the hull fills
Port rod, Starboard rod	Rails	Cast, set, fight, land (see Fishing)	Rigged rods can sit in holders; a strike without a hand usually breaks off
Stern rods (2)	Stern holders	Trolling rods for the whole night	They fish alone but only hold small fish
Net winch	Stern gantry	Shoot and haul the trawl	A shot net keeps fishing and keeps getting heavier
Lantern mast	Midships	Raise, lower, hood, or colour the lamp	Stays where it was left
Sonar and radio	Wheelhouse	Pings, reads contacts, radios other stations and the Owners	Nobody sees below the surface
Harpoon cannon	Bow	Aim, fire, reload (reloading is faster with two)	—
Gutting table and ice hold	Aft of the wheelhouse	Gut, grade, and ice fish so they keep their value	Fish on deck lose freshness and bleed
Air pump	Beside the diving gantry	Keeps a diver breathing (see Diving)	The diver has 30 s of air in the helmet

Suggested six-hand watch. Helm and sonar (one person, swapping), boiler and pumps, two rods, net and gutting, lantern and cannon. When a dive goes down, one rod hand moves to the air pump. When something big comes, the rod hands drop everything for the cannon and the gaffs.

Moving on the deck. Top-down, 8-way movement at 3 m/s, slower carrying (a 40 kg tuna halves speed; two hands carrying one halves it less). Ladders to the wheelhouse roof, the mast, and the engine room. The deck is wet: at more than 12° of roll, unbraced players slide; at 25°, they fall. Holding a rail or a station braces.

Inventory. Four slots per hand (Lethal Company's rule), one of them the hands. Large fish and salvage are carried in both hands and can be thrown short distances. Tools: gaff, knife, lamp, flare pistol, speargun, rope, and so on.

Talking. Communication is the core skill, so the game carries it, not a third-party app.

- **Proximity voice** through the shared arcade layer, the game's main channel: voices carry about 12 m on deck and are muffled by walls and wind. Heavy rain and a running engine cut the range in half.
- **Speaking tube** from the wheelhouse to the engine room and to the bow, always clear.
- **Walkie** (bought): channel-wide voice, one per slot, runs on batteries.
- **Ping wheel** for players without a microphone: "Fish on," "Need gaff," "Steer port," "Something below," "Get out of the water," and a point marker.
- **The ship's bell:** anyone can ring it; everybody hears it anywhere on the boat, and the dead can ring it too.

Controls, HUD, and the sonar

The HUD is almost empty on purpose: a player learns the sea from what their station shows them and what the crew says. There is no health bar, no minimap, and no enemy markers.

Controls (keyboard and mouse)

Action	Input
Move	WASD
Brace, hold a rail	Hold Shift
Take a station, pick up, use	E
Leave a station	X
Primary (cast, fire, reel, pump, shovel)	Left mouse

Action	Input
Secondary (aim a cast, sight a weapon, gaff)	Right mouse
Rod angle, aim	Mouse direction
Drag, lantern level, telegraph setting	Scroll
Set the hook	Space
Inventory slots, drop	1-4, Q
Hand lamp	F
Reload	R
Push-to-talk (open mic is an option)	V
Walkie	T
Ping wheel	Hold G
Crew board (roles, injuries, who is dead)	Hold Tab

A full gamepad map (triggers for reel and cast, right stick for rod angle) ships with the first playable build; every binding is remappable.

What each station shows

Station	Overlay
Rod	The reel gauge: a tension arc with the drag setting marked, a line counter (metres out), and the lure's depth
Helm	Telegraph, heading, speed, and a chart inset of the ground with set gear buoys
Boiler	Pressure gauge and coal in the bunker
Bilge pumps	Water level in each flooded section
Net winch	Warp tension and estimated load
Lantern mast	Level and filter colour

Station	Overlay
Harpoon cannon	A reticle with drop and the refraction offset shown as a faint second ring
Air pump	The pump rhythm bar, the diver's air, and hose tension
Gutting table	Species name, weight, grade, and freshness of the fish in hand

Always on screen: the four inventory slots and carried weight (bottom left), a ping marker when a crewmate points something out, and the ship's bell icon when it rings. **Time** is read from the wheelhouse clock or a bought pocket watch, as in *Lethal Company*. **Injuries** show as small icons on the crew board and as their effects (a broken arm hangs; a bleeding player leaves drops). **Overboard** shows as a narrowing breathing vignette with the time left.

The sonar

The wheelhouse sonar is a round phosphor scope (the same visual language as the expedition sonar chart) and the crew's only view below the surface outside the lantern's light.

- **Passive** (always on): shows only loud things near the boat. At half speed or more, the *Gannet*'s own screw drowns it out.
- **Active ping:** one every 3 s, 150 m radius. Returns show for 6 s and fade. Every ping writes noise into the water at the boat, which Ghost Worms and curious species hear.
- **Depth dial:** the operator picks which depth band the scope shows; a side-scan toggle shows the water column under the boat as a profile.
- **What returns look like:** schools as dotted clouds, single fish as blips sized by size class, set gear as bright squares, wrecks as hard shapes, kelp as shadow that hides what is behind it, and in the Grotto every return has a faint echo twin. The Kraken whites out the whole scope.
- **Marking:** the operator clicks a contact to mark it; every crewman sees a bearing arrow at the edge of their screen for 10 s.
- **Second sonar** (Slipway upgrade): a second operator, and depth numbers on every contact.

Dock screens

The dock is walked, not menued: the chalkboard (quota, nights, money, role upgrades), the Chandler's counter, the Fish Market scales (fish are carried and weighed one by one, or in crates), the Owners' office window (salvage), the Slipway, and the chart table on board.

Boat and sea physics

The boat, the water, every line, and every fish on a line are one physical system stepped at 60 Hz on the host (`trawl_boat.cpp`). The rule: anything that pulls on the boat moves the boat, and anything that moves the boat moves the crew.

The sea surface. Height at any point is a sum of four directional waves (a cheap Gerstner set) whose amplitude comes from the weather. The top-down view shows swell as moving light bands and foam; the side views (dive, reel camera) show the actual profile. A per-ground current vector and the wind vector push everything on the surface: the boat, floats, the net, corpses, and blood.

The hull. A rigid body in the plane (x, y, heading) plus three scalar degrees of freedom: roll, pitch, and heave. Buoyancy is computed from eight hull sample points against the wave height. Weight on board is not a single number: the hold, the ice, every fish on deck, every player, the net drum, and water in the bilge each sit at a position, so a full net on the stern pitches the bow up and five crew at the port rail during a fight list the boat to port. Roll above 12° makes loose items and unbraced players slide; above 35° the boat is on its beam and takes water over the rail.

Damage. The hull has six sections, each with integrity 0–100. Rams, bites, rocks, and grinding against a wreck lower it; below 40 a section leaks, below 10 it floods fast. Leaks fill the bilge; the pumps empty it. Hull plates (bought) raise section maximums; a hand with a patch kit can stop a leak in 6 s while standing in water.

The engine. Coal into the firebox raises boiler pressure; the telegraph setting draws pressure into shaft speed. Pressure in the green gives full thrust; red for more than 5 s blows a relief valve (a steam jet that burns anyone in the engine room and stops the screw for 20 s). The screw is loud: the engine's noise output feeds the ecosystem's sound layer by telegraph setting (stop 0, slow 2, half 5, full 9).

Lines, nets, and ropes. Every line (rod lines, the harpoon tether, the trawl warps, the diver's air hose, the life-ring rope, the anchor) is a Verlet chain with 8–24 segments, reusing the `Chain` code from the master reference. Tension is read from the constraint

error at the anchor end. When tension exceeds the line’s rated strength for longer than its shock window (0.2 s for mono, 0.5 s for braid, 1 s for steel), it snaps. Lines can foul each other: two rod lines crossing under the boat tangle and must be cut or walked around.

Light. The lantern mast, deck lamps, hand lamps, flares, and bioluminescence write into a light grid shared with the ecosystem (a new light layer, see The food web). The top-down view is dark except where light reaches; the sea beyond the lantern radius is drawn black with only foam and glints. Lantern levels: hooded (4 m), low (8 m), full (14 m), searchlight (a 30 m cone the operator aims).

Weather. Rolled per night and ground, changing once or twice a night.

Weather	Swell	Effect
Calm	0.2 m	Voice carries far; fish bite light
Fog	0.3 m	Light radius –40%; sonar is the only way to see anything coming
Rain	0.6 m	Voice range halved; surface feeders bite more
Squall	1.5 m	Roll up to 20°; lines tangle; big predators hunt the surface
Storm	3 m	Roll up to 30°; waves wash the deck; only the brave fish
The Glass	0 m, mirror-flat, no wind	Rare. Everything stops biting. It precedes something enormous

Moon. Full moon doubles plankton at the surface (more bait fish, more predators up high); new moon brings deep species shallower. The phase advances one step per night, so the three nights of a deadline differ.

Fishing

Fishing is a physics fight between a player, a rod, a line, and an animal that is also an agent in the food web. The UI shows only what an angler would feel: rod bend, line angle, reel sound, and a tension gauge on the reel. There is no minigame bar.

Tackle

Tackle	Line strength	Cast range	Reel speed	Holds	Price
Handline	8 kg	6 m (thrown)	Hand over hand	Bait fish, small reef fish	Free
Light rod	6 kg	30 m	Fast	Surface and reef fish to 5 kg	40
Medium rod	15 kg	25 m	Medium	Most fish to 25 kg	120
Heavy boat rod	40 kg	15 m	Slow, 2-speed	Tuna, sharks, groupers	300
Deep-drop reel	30 kg	Drops straight down, 200 m	Steam-assisted	Deep species in the Trench and Atlantis	450
Big-game chair	120 kg	15 m	Slow, 2-speed, fighting harness	Marlin, giant tuna, small Ghost Worms	900 (boat upgrade)

Line types: mono (stretchy, forgiving, visible to wary fish), braid (no stretch, strong, cut by teeth and coral), wire leader (bite-proof, spooks wary fish), glow line (seen by everything). Hooks: small, circle, big-game treble. Stats live in `trawl_data.cpp`.

Bait and lures

Every species has a diet in the food web, so bait is the food web in the player's hands. A fish takes what it eats.

- **Dead bait:** squid strips, shrimp, chunks cut from caught fish (a tuna chunk catches sharks). Loses scent over 3 minutes.
- **Live bait:** bait fish caught on handlines and kept in the bait well. Swims on the hook, draws predators by movement and by scent, and can itself be eaten by something bigger than what you wanted.
- **Lures:** spoon (flash, mid-water), jig (vertical, deep), popper (surface, loud), glow lure (light), rattle (sound), plug (dives to a set depth). Each has a flash, noise, and scent profile the species' senses score.
- **Chum:** offal from the gutting table or a bought chum bucket. Writes blood into the scent grid. Brings fish fast, and brings whatever eats them.

The cast

Hold to charge, release to cast; aim with the mouse. Wind drifts the lure, and a cross-wind cast lands off target. The lure sinks at its own rate; the angler counts depth by the line counter on the reel. Retrieve style is the player's input: steady reel, jig (tap), twitch-and-pause, or dead drift. Each species scores the presentation against its preference; a matched presentation roughly triples the bite chance.

The bite

A fish that decides to take goes through **inspect** (the rod tip ticks), **nibble** (small taps, 1–4), and **take** (the rod loads). Setting the hook during the take (a 250 ms window, 150 ms for wary species) hooks it; striking on a nibble spooks it and teaches that school to avoid that lure for 2 minutes. Circle hooks set themselves if the player simply reels.

The fight

The hooked fish is an agent with **strength**, **stamina**, and a **fight pattern** from its species record: runs, deep dives, jumps and head-shakes, circling under the boat, sounding into cover (a wreck or kelp, where the line chafes), or rolling (sharks, which wrap the line).

The angler controls three things:

1. **Reel** (hold): gains line only when tension is below drag.
2. **Rod angle** (mouse direction relative to the fish): pulling against the direction of the run tires the fish fastest (side pressure); pulling straight back wastes stamina; a high rod angle while it jumps throws the hook.
3. **Drag** (scroll): how much tension slips line instead of breaking it. Low drag is safe but lets the fish run into cover; high drag ends fights fast and snaps lines.

Tension is computed from the chain physics, not a formula for the UI. It breaks lines over their rating; below 5% for 1.5 s (slack) lets barbless or circle hooks fall out. A fish's pull is a real force on the boat: a 200 kg marlin can tow the *Gannet* at slow speed, and the helm can help by backing the boat down on it.

Fight tuning

Every hooked fish runs one model, shaped by its fight pattern. All constants live in `trawl_data.cpp`; these are the starting values `--trawl-fight` is tuned against.

The model

- **Pull:** a fish's pull in kilograms-force = its weight × the pattern's pull ratio, rising to 1.5× during a burst.
- **Stamina:** seconds of full pull = $20 + 6 \times \sqrt{\text{weight}}$ (a 4 kg snapper has 32 s; a 200 kg marlin has 105 s).
- **Drain:** stamina drops by (line tension ÷ pull) per second. Side pressure (rod angle within 45° of square to the fish's heading, pulling against its turn) multiplies drain by 1.5; pulling straight back by 0.6; slack drains nothing.
- **Recovery:** when tension falls under 20% of pull, the fish recovers 0.3 s of stamina per second, so a timid angler never ends a fight.
- **Landing:** at under 15% stamina the fish comes alongside and can be gaffed, netted, or lifted; it bursts once more if the gaff misses.
- **Passing the rod:** a crewman can take over a rod with E; the handover gives 1 s of held tension, no slack.

Fight patterns

Pattern	Pull ratio	What it does	What the angler must do
Run	0.8	Bursts of 4–8 s at up to 3 m/s every 10–15 s	Let drag take the run, then side pressure between runs
Dive	0.9, straight down	Sounds for depth; each 10 m gained adds 5% to its pull	Pump: lift the rod, then reel down as it drops; the helm can move over it
Jump	0.6	Leaps every 8–12 s	Lower the rod (bow) within 0.3 s of the jump or a 40% chance to throw the hook
Circle	0.5, steady	Circles under the boat	Keep the line off the keel and other lines; a keel pass chafes the line 10%
Cover	0.7	Heads for the nearest cover within 15 m	High drag to turn it early; in cover the line chafes 5% of its strength per second
Roll	0.6	Rolls to wrap the line	Wire leader, or each wrap cuts effective line strength by 10%
None	—	Lifted straight in	—

Target fights (sensible bot angler, matched tackle)

Fish	Tackle	Fight time	Landed
3 kg snapper	Light rod	20–40 s	85%
15 kg lingcod	Medium rod	60–120 s	75%
40 kg yellowfin	Heavy rod	3–5 min	60%
150 kg sturgeon	Deep-drop reel	10–20 min	45%
250 kg blue marlin	Big-game chair	8–12 min	40%

Fights longer than these write more vibration into the water, so the model and the food web push the same way: a long fight is a dangerous fight.

Landing

- Fish under 5 kg: lifted over the rail.
- 5–40 kg: a second hand with the gaff or the landing net. Gaffing has a timing window as the fish comes alongside; a miss spooks it into one more run.
- 40 kg and over: tail rope, then the net winch or the cannon's winch hauls it aboard, and the whole boat lists while it comes up.

A landed fish flops on deck (a physics body that slides with the roll and can flop back overboard) until someone takes it to the gutting table.

The trawl

The net is the big-money, big-risk tool. Shooting it takes 15 s; while down it is a Verlet mesh of 8×6 nodes towed on two warps. It fills by sweeping through schools (catch rate = school density $\times$ mouth width $\times$ speed), gets heavier, and drags the stern down and slows the boat. It catches everything in its path, including what the crew didn't want: bycatch jellies sting whoever empties the net, and a sleeping Ghost Worm in the mouth turns the haul into a fight. It snags on wrecks and reefs (warps shriek, the boat slews) and must be backed off or cut loose, losing the net. Hauling needs a hand on the winch and a hand guiding the boom. A full net's cod end is emptied onto the sorting deck in one wet, flopping pile.

Set gear

- **Longline:** 20 baited hooks on a line with a radio buoy at each end, set and left to fish. Returning finds fish, empty hooks, or fish heads: other predators strip hooked catch (depredation), which tells you what is feeding there.
- **Pots:** crab and lobster pots dropped on a reef or wreck with a float, hauled on a later pass.
- Set gear drifts with the current and can be stolen by the Ghost Ship's crew.

Shooting

Guns are the fast, loud, bloody way to take fish and the only way to stop some threats. Every shot trades value and safety for speed: a shot fish is worth less than a hooked one, it bleeds, and the report carries through the water.

Ballistics. Projectiles are simulated, not hitscan. Above the surface: gravity and light drag. At the surface: a round's velocity drops to 10% in the first 0.3 m and it stops within 1.5 m, so rifles only work on fish at or breaking the surface. Harpoons and spears keep their energy far longer in water. Refraction: a target seen below the surface is drawn where the light says it is, 15–25% nearer and shallower than it really is, so aiming at a fish in the water means aiming below it. Hits resolve against the fish's body (head, body, tail); a head hit on a fish under its size class kills clean.

Weapons

Weapon	Where	Range	Use	Noise	Price
Gaff	Hand	Melee	Landing fish; beating off boarders	1	Free
Fish priest (club)	Hand	Melee	Kills a landed fish instantly, stops flopping	1	Free
Speargun	Hand, tethered	8 m in water, 12 m in air	Taking fish near the hull; divers' weapon	2	150
Flare pistol	Hand	40 m arc	Light burst (scares light-shy species, draws curious ones); signals	3	60 + 10/flare
Rifle	Hand	Surface and air	Breaching fish, gulls, boarders, the Siren's heads	6	250 + 2/round
Shotgun	Hand	15 m, air	Gull flocks, boarders	7	200 + 3/shell

Weapon	Where	Range	Use	Noise	Price
Harpoon cannon	Bow mount	35 m, surface to 6 m deep	Big fish and big threats; the tether goes to its winch	5	700 (boat upgrade)
Explosive harpoon head	Cannon ammo	—	Kills, but destroys most of the fish's value	9	80 each
Depth charge	Thrown or rolled off the stern	12 m blast radius	Stuns everything in range, floating fish to the surface	10	120 each

The tethered harpoon. A harpoon hit makes the target a hooked fish (see Fishing) with the cannon's winch as the reel and 120 kg steel as the line. This is how the crew lands a Great White or holds a Ghost Worm's head up while the rifles work. A tethered giant tows the boat; the winch can be released in an emergency.

Shot fish and value. Clean hook catch 100% of value, speared 85%, rifle-shot 70%, harpooned 75% (big fish), explosive harpoon 30%, depth-charged 50% (and every stunned fish floats: the charge takes everything, including species worth nothing). A wounded fish that escapes bleeds into the scent grid as it goes.

Noise and blood. Each shot writes its noise value into the sound layer at the muzzle and again at the impact point. Fearful species scatter; curious species come; aggressive ones come fast. Every wound writes blood ($\text{size} \times \text{wound fraction} \times 2$ per second, $\text{size} \times 20$ on death). A depth charge is a dinner bell for a whole ground: it is the fastest money in the game and the fastest way to raise Wake (see The food web).

Friendly fire is on. A rifle round or a harpoon can hit a crewman, and a harpoon tether can drag one over the rail.

The food web

Every ground runs one ecosystem simulation on the engine code Red Tide already has (agents, the scent and sound grids, the species file loader). The Trawl reuses that code to save time, not its design: the rules, numbers, species, and tuning here are written for a boat on the surface at night. The Trawl's web has seven layers (diet, blood and scent, sound, territory, symbiosis, flora, and the crew) plus three that exist only in this game: light, vibration, and Wake. The crew is a species in the web, so everything they catch, spill, and shine changes what comes next.

Every catch feeds something bigger, and the crew is on the menu



Arrows run from food to eater. What the crew spills raises Wake, and Wake calls the apex to the boat.

The Trawl food web · five trophic bands, the crew, and Wake

What each ground holds

About 25 fish and invertebrate species, 6 threats, and 4–6 flora per ground, as records in `trawl_species.json` using the shared species file format (class, size 1–6, trophic tier, diet, blood yield and threshold, senses, temperament, home, social). The Trawl adds four fields: **depth band** (surface, upper, mid, deep, abyss, with a night shift: many species rise after dark), **light response** (drawn, neutral, or shy, by colour), **bite profile** (preferred bait, lure flash, noise, and retrieve), and **fight pattern** (runs, dives, jumps, circles, sounds into cover, rolls).

Red Tide: shared assets and code, separate games. Red Tide and the Trawl are different games. They share only what saves time: the ecosystem engine code, the species file loader, and, where the same animal appears in both (a Great White, a barracuda), its sprite sheet and sound cues as a starting point to re-colour and re-pose. Nothing about design carries over. Every Trawl species has its own record, numbers, and behaviour written for this game: fish are animals you fight on a line and read from above through sonar and light, not beasts a diver shoots.

The three new layers

1. **Light.** The light grid from the boat physics feeds the ecosystem. Plankton and forage fish rise toward light; predators follow the forage; light-shy deep species retreat from it; curious threats (Anglers, Sirens, Drowned sailors) are drawn to it. A full lantern roughly doubles bites after 30 minutes by pulling a whole food chain up under the hull. Red light is invisible to most deep species.
2. **Vibration.** Taut lines, the trawl, the screw, a thrashing hooked fish, and hull impacts write vibration into the water. Ghost Worms, sharks, and anything with a lateral-line sense track it. A fish fought for five minutes is a beacon.
3. **Wake.** A per-ground scalar (0–100) that sums what the crew has put into the water: blood above the ambient level, loud noise, bright light, cursed salvage, the Pale Eye's gaze, and depth charges (+15 each). It decays 1 point a minute when the crew is quiet and dark. Wake is what calls the big three (see Threats) and it is shown to the crew only indirectly: the music, the fish going quiet, the sonar's clutter, and the ghosts' outlines.

How the web plays

- **Bait chains.** Handline silversides, live-bait them for mahi, cut the mahi for shark chunks. Each step up the chain is worth more and bleeds more.
- **Depredation.** A hooked fish is prey. A tuna on a line for two minutes in the Weeds has a real chance of coming up as a head; a barracuda pack strips a longline. What took the fish tells the crew what is under them.
- **Satiation.** A predator that has just eaten ignores bait for a while. A Great White that took a tuna off the line gives the crew a few quiet minutes; a hungry one follows the boat.
- **Schools.** Forage fish move as boids; predators herd them into bait balls near the surface, which the sonar shows and which surface feeders and gulls give away. Netting through a bait ball is the best trawl of the night, and every predator around it is already there.
- **Cascades across nights.** Because a ground remembers across a deadline, heavy netting of forage fish on night one leaves predators hungry on night two (more bites, more bold approaches, more depredation), and killing the Weeds' sea otters lets urchins strip the kelp, emptying the canopy by night three.
- **Bycatch and discards.** Everything the net brings up that isn't sold goes back over the side dead, as blood and food, feeding scavengers and raising Wake. Sorting quickly and returning bycatch alive (a hand at the sorting chute) keeps the water quieter.

- **Blood rules.** Blood diffuses to neighbouring cells each second and decays 3% per second in open water (1% in the Grotto's still water) with the current carrying it downstream, so a slick trails away from the boat and predators arrive from downstream.

Simulation budget

`--trawl-eco <ground> <minutes> [pattern]` runs the web headless with a scripted crew and prints populations, blood, Wake, and threat arrivals. A ground with no crew must stay stable (no species extinct and none above 150% of its start) for three simulated nights.

The four grounds

The four grounds are Depth's four expedition locations seen from the surface, from safest to deadliest. Each is a map about 600 m across with its own depth chart, catch table, flora, threat roster, and one rule no other ground has.

Ground	Depth	Coal to reach	Fish value	Salvage	Signature rule	Starting-gear quota rate (target)
Eclipse Lagoon	3–40 m	10	Low	Little	The reef tide	85%
The Weeds	8–70 m	25	Medium	Some	Kelp fouls everything	65%
The Grotto	5–120 m, under rock	40	Medium-high	High (wrecks)	The cave mouth closes	45%
Atlantis Waters	30–400 m	80	Very high	Very high (relics)	The Pale Eye	25%

Eclipse Lagoon (the Island)

A turquoise atoll lagoon under the Island's eclipse moon, with a coral crest between the lagoon and the open sea. It is where crews learn to fish and learn that blood matters.

- **Catch.** Bait: silversides, sardines, flying fish. Reef: snapper, parrotfish, triggerfish, spiny lobster (pots), octopus. Open-water edge: mahi, bonito, barracuda, grouper in the crest's caves, reef sharks.
- **Flora.** Coral heads (shelter, snag nets), seagrass (grazers), sargassum rafts (drift with the wind; mahi shelter under them).

- **Signature: the reef tide.** The tide falls through the night. After 01:00, parts of the crest are too shallow for the *Gannet*; grounding holds her fast until a hand rows a kedge anchor out and the winch pulls her off.
- **Threats.** Gull flocks (steal fish off the deck), barracuda packs, reef sharks, Island war canoes on some nights (they trade fish for gear, or raid for it, depending on how the crew answers their drum; ties to Fathoms' Islanders).

The Weeds

A kelp forest whose canopy reaches the surface in dark golden mats, with clear lanes between. Everything big lives at the edges.

- **Catch.** Bait: anchovy balls, kelp perch. Mid: kelp bass, sheephead, rockfish, bonito. Edge runs: bluefin tuna (a 20-minute fight on the heavy rod), yellowtail. Bottom: halibut, abalone (by diving).
- **Flora.** Kelp (shelters fish, blocks sonar, fouls lines), urchin barrens (where kelp has been grazed away; empty and exposed), sea otter rafts that eat urchins and keep the kelp standing.
- **Signature: kelp fouls everything.** Running the engine through the canopy wraps the screw (speed halves until a hand cuts it free from the stern ladder, in the water). Nets fill with kelp that weighs as much as fish and is worth nothing. Hooked fish run into the canopy to chafe the line.
- **Threats.** Sirens (sing from the kelp lanes, pull the helm toward the rocks), Kelp Wraiths (drift unseen in the canopy, entangle a crewman at the rail), a Great White that patrols the seaward edge, Feral Mermen who cut nets.

The Grotto (the Cave)

The *Gannet* sails through a sea arch into a flooded cave the size of a cathedral: black water, dripping stalactites, glowing mould on the walls, and the hulls of smugglers' ships that went in and stayed.

- **Catch.** Blind cave fish, cave shrimp (bait), glass eels, lantern fish, cave cod, giant isopods (pots), the Grotto sturgeon (big, ancient, worth a great deal).
- **Flora.** Glowing mould (light source; grazers feed on it), pale weed, bacterial mats around warm vents.
- **Signature: the cave mouth closes.** The arch is only passable while the tide is low enough. The telegraph posts the closing time at the start of the night (between 02:30 and 04:00). A crew still inside when it closes is shut in until 05:00, which the Owners

count as late, and the cave's night keeps waking the whole time.

- **Echo.** Sound doubles in the Grotto: every ping, shot, and shout feeds the sound layer at twice its value and returns as an echo on the sonar. Loud noises can drop stalactites (hull damage, crew damage).
- **Threats.** Lantern Anglers (the second light), Ghost Worms (born in the Grotto), Drowned sailors from the wrecks, swarms of isopods that climb the anchor chain.

Atlantis Waters

Open ocean above the drowned city. The water drops from the terraces at 30 m to the Trench's edge at 400 m. The Pale Eye watches from the deep; the richest fishing in the game and the best relics sit under it.

- **Catch.** Surface: marlin, giant bluefin, flying squid. Mid: swordfish, opah, oarfish. Deep-drop: Atlantean coelacanth, gulper eels, the drowned-gold grouper. Everything here is large, and a single marlin can cover a night's quota share.
- **Flora.** Sunken gardens on the terraces, deep coral, the Eye's bioluminescent veils (lights that drift up through the night).
- **Signature: the Pale Eye.** A great pale light far below. While any crewman looks at it (the searchlight pointed down, a diver facing it, or anyone on the sonar with a ping open over it), Wake rises twice as fast. It blinks when something big is coming.
- **Threats.** Everything from the other grounds plus the Kraken, the Ghost Ship, the Deep Choir (a Siren chorus that sings the whole crew), and Cult longboats that chum the water around the *Gannet* on purpose.

Rumors and variants. Each ground rolls one variant per night (a sardine run, a red tide bloom that kills bait fish, a whale carcass that draws every scavenger, a wreck exposed by a storm, a school of tuna at the surface, a missing boat's lights). The telegraph's rumor hints at it, correctly 70% of the time.

Nightly variants

Each night rolls at most one variant for the chosen ground; about 40% of nights roll none. The telegraph's rumor points at the variant (right 70% of the time). Every variant changes the web, not just a number, so the crew has to read it.

Ground	Variant	Chance	What changes
Any	Bait run	8%	Forage fish ×3 near the surface; every predator ×1.5 and bolder by midnight
Any	Whale carcass	6%	A floating carcass leaks blood all night; scavengers, sharks, and the ground's apex gather on it; oil and ambergris salvage (150–400) for anyone who cuts in
Any	Red tide bloom	6%	Forage fish die and float; predators are hungry and take any bait; fish caught near the bloom sell at 50%
Any	Drifting derelict	5%	A dark boat drifts through; boarding it is a small salvage run with something still aboard
Any	Storm-exposed wreck	5%	An extra shallow wreck to dive, sonar shows it at the start
Eclipse Lagoon	King tide	10%	The crest stays passable all night; open-sea species come into the lagoon
Eclipse Lagoon	Turtle nesting	8%	Turtles everywhere; net bycatch fines add up fast; tiger sharks follow the turtles
Eclipse Lagoon	Canoe night	8%	Island war canoes are out: trade, pay tribute, or fight
The Weeds	Tuna run	10%	Bluefin along the seaward edge from 22:00; the Great White follows at Wake 30 instead of 40
The Weeds	Kelp storm	8%	Drift mats everywhere; screw fouling twice as often; yellowtail under every mat
The Weeds	Mermen's market	6%	Feral Mermen trade abalone and relics for fish instead of cutting nets
The Grotto	Glass eel run	10%	Glass eels pour in with the tide; pale cod and anglers follow them to the light

Ground	Variant	Chance	What changes
The Grotto	Rockfall	8%	A new chamber opens with an untouched wreck; stalactites fall at half the usual noise
The Grotto	Mould bloom	8%	The cave glows; everything bites, Anglers stay away, but Drowned sailors see the boat from anywhere
Atlantis Waters	The Eye wide open	10%	Wake builds 1.5× all night; relics on the terraces glow and show on the sonar
Atlantis Waters	Swordfish night	8%	Swordfish at the surface from dusk; the sperm whale and a giant squid are both present
Atlantis Waters	Procession	6%	Cult longboats from the start of the night, chumming a ring around a sunken altar full of relics

Species rosters

Each ground's catchable species, as they go into `trawl_species.json`. Threats have their own stat blocks (see Threats). Tier: 1 forage or grazer, 2 small predator, 3 mid predator, 4 apex. Depth bands: surface 0–2 m, upper 2–15, mid 15–60, deep 60–200, abyss 200+. Price is the Fish Market base in shillings per kilogram before grade, freshness, and glut. Fight patterns are tuned in Fishing: Fight tuning.

Eclipse Lagoon

Species	Tier	Weight (kg)	Depth	Eats	Takes	Fight	Price (sh/kg)
Silverside	1	0.05	Surface	Plankton	Handline, glow speck	None	0.5 (bait)
Sardine	1	0.1	Surface-upper	Plankton	Handline, net	None	0.8
Flying fish	1	0.3	Surface	Plankton	Net; glides onto a lit deck	Jump	1.5
Reef squid	2	0.3–1	Upper	Silversides	Glow jig	Jump (inks)	4

Species	Tier	Weight (kg)	Depth	Eats	Takes	Fight	Price (sh/kg)
Surgeonfish	1	0.5–1.5	Upper, reef	Algae	Shrimp	Circle	2
Parrotfish	1	2–5	Upper, reef	Coral algae	Shrimp, small jig	Cover	2
Grunt	2	0.5–2	Upper	Shrimp, worms	Shrimp	Circle	1.5
Triggerfish	2	1–4	Upper, reef	Urchins, crabs	Squid strip	Cover	3
Blue crab	2	0.3–0.8	Seagrass floor	Carrion	Pots	None	5
Spiny lobster	2	1–4	Reef floor	Urchins, carrion	Pots	None	8
Reef octopus	2	1–6	Reef holes	Crabs, lobster	Crab bait, pots	Cover (inks)	6
Snapper	2	1–6	Upper–mid	Small fish, shrimp	Shrimp, squid, spoon	Run	3
Bonito	3	2–6	Surface	Sardines	Spoon, feather	Run	3
Jack crevalle	3	5–15	Upper	Sardines	Popper, spoon	Circle	1.5
Barracuda	3	3–20	Upper	Sardines, snapper	Spoon, live bait	Run, jump	1.5
Moray eel	3	3–15	Reef holes	Fish, octopus	Dead bait at night	Cover	1
Mahi-mahi	3	5–20	Surface, under sargassum	Flying fish, squid	Live bait, popper	Jump	4
Grouper	3	8–40	Crest caves	Lobster, octopus, fish	Live bait, jig	Cover	5
Reef shark	4	20–60	Upper–mid	Fish	Chunks on wire	Roll	2
Green turtle	2	40–120	Upper	Seagrass, jellies	Bycatch only	Dive	Protected
Moon jelly	1	—	Surface	Plankton	Net bycatch	None	0 (stings)

Flora: coral heads (shelter, snag nets), seagrass (grazer food, crab habitat), sargassum rafts (drift with the wind; mahi and flying fish shelter under them), plankton bloom (rises under light and a full moon).

Web notes. Parrotfish keep algae off the coral; netting them out lets algae smother coral heads over three nights, and the reef fish that shelter there thin out. A landed turtle costs a 50-shilling Owners' fine unless returned alive within 60 s. Barracuda sell cheap but are the Lagoon's main thieves of hooked fish.

The Weeds

Species	Tier	Weight (kg)	Depth	Eats	Takes	Fight	Price (sh/kg)
Anchovy	1	0.03	Upper	Plankton	Net, handline	None	0.6
Market squid	1	0.1	Surface at night	Plankton, anchovy	Light and net, glow jig	None	2
Kelp perch	1	0.2	Upper, canopy	Plankton, isopods	Tiny hook	None	0.8 (bait)
Opaleye	1	1–3	Upper, canopy	Kelp	Green-weed bait	Cover	2
Purple urchin	1	0.2	Rock floor	Kelp	Diving	None	3 (roe)
Abalone	1	0.5–2	Rock floor	Kelp	Diving only	None	12
Kelp crab	1	0.3	Canopy	Kelp	Pots	None	4
Rock crab	2	0.5–1.5	Floor	Carrion	Pots	None	6
Pacific mackerel	2	0.5–1.5	Upper	Anchovy	Spoon, feather	Run	1.5 (best live bait)
Kelp bass	2	1–6	Upper–mid, canopy	Perch, crabs	Swimbait, live perch	Cover	3
Sheephead	2	2–10	Mid, canopy	Urchins, crabs	Crab bait	Cover	3
Vermilion rockfish	2	1–5	Mid–deep	Small fish	Jig	Dive	3.5
Bat ray	2	5–30	Sand floor	Clams, crabs	Squid on the floor	Run (low, heavy)	0.5
Leopard shark	3	5–20	Floor	Crabs, worms	Squid	Roll	1
Lingcod	3	5–25	Deep rocks	Rockfish, octopus	Big jig, live bait	Dive, cover	4
Halibut	3	5–40	Sand floor	Fish	Drifted live bait	Dive	5

Species	Tier	Weight (kg)	Depth	Eats	Takes	Fight	Price (sh/kg)
Yellowtail	3	5–20	Upper, seaward edge	Mackerel, squid	Live mackerel, iron jig	Run, cover	4.5
Giant sea bass	4	50–250	Deep kelp edge	Fish, crabs, rays	Live mackerel, heavy rod	Dive, cover	3
Bluefin tuna	4	40–200	Surface–mid, seaward edge	Mackerel, anchovy, squid	Live mackerel, popper	Run, circle	7
Sea otter	2	15–40	Surface rafts	Urchins, abalone, crabs	Never	—	Protected

Flora: giant kelp (canopy shelter; blocks sonar; fouls lines, nets, and the screw), eelgrass beds (nursery for perch and crabs), urchin barrens (bare rock where kelp has been grazed away), drift kelp mats (float with the current; yellowtail shelter under them).

Web notes. The Weeds are the game's clearest cascade: otters eat urchins, urchins eat kelp, kelp shelters everything else. Killing or scaring off an otter raft (gunfire near them counts) lets urchins spread, turning a kelp patch into barrens over the three nights of a deadline, and the bass, perch, and sheephead that lived there leave. An otter killed costs an 80-shilling fine. Bluefin only run along the seaward edge, where the Great White patrols.

The Grotto

Species	Tier	Weight (kg)	Depth	Eats	Takes	Fight	Price (sh/kg)
Cave shrimp	1	0.02	All bands	Mould film	Net, shrimp trap	None	1 (bait; glows)
Glass eel	1	0.05	Upper	Plankton	Net under a light	None	6
Lantern fish	1	0.05	Mid-deep	Plankton	Rises to light; net	None	1.5
Blind cavefish	1	0.1–0.3	Mid	Shrimp	Scented bait (it cannot see lures)	None	4

Species	Tier	Weight (kg)	Depth	Eats	Takes	Fight	Price (sh/kg)
Mould snail	1	0.2	Walls	Glowing mould	Diving	None	2
Pale cave crab	1	0.3	Floor	Mould, carrion	Pots	None	5
Hagfish	1	0.5-1	Floor	Carrion	Any dead bait on the floor	Roll (slimes the line)	1
Giant isopod	2	1-3	Floor	Carrion	Pots	None	7
Bone crab	2	1-3	Wrecks	Carrion	Pots set in wrecks	None	6
Albino octopus	2	1-5	Wrecks	Crabs	Crab bait	Cover (inks)	7
Echo bass	2	1-4	Upper	Shrimp	Rattle lure (hunts by sound)	Circle	4
Pale cod	2	2-8	Mid	Shrimp, cavefish	Glow jig, shrimp	Dive	4
Cave dogfish	3	5-15	Deep	Cod, eels	Chunks	Roll	2
White conger	3	5-25	Walls, wrecks	Cod, crabs	Dead bait on the bottom	Cover, roll	3
Ghost ray	3	10-40	Floor	Crabs, isopods	Squid on the floor	Run (glides)	3
Grotto sturgeon	4	60-250	Deep floor	Crabs, cod	Big dead bait, deep-drop reel	Dive (slow, very long)	8, plus 200 for a female's roe
Glowjelly	1	—	Upper	Plankton	Net bycatch	None	0 (stings; a light source)

Flora: glowing mould (the Grotto's main light and its base food), pale weed (shallow shelves), bacterial mats around warm vents (deep; draw isopods and hagfish).

Web notes. The Grotto's web runs on mould, not sunlight, so it is small and fragile: a heavy night of netting shrimp starves the cod and cavefish by night two, and hungry cod bring the Lantern Anglers up. Sound doubles here (see The Grotto), so echo bass come to noise that sends everything else running. Sturgeon take 10-20 minutes on the deep-drop reel, and the vibration of that fight is the most reliable way to wake a Ghost Worm.

Atlantis Waters

Species	Tier	Weight (kg)	Depth	Eats	Takes	Fight	Price (sh/kg)
Lantern fish	1	0.05	Deep by day, upper at night	Plankton	Rises to light; net	None	1.5
Mackerel scad	1	0.2	Upper	Plankton	Sabiki rig	None	1 (bait)
Barreleye	1	0.1	Deep	Jellies	Net	None	30 (a curiosity)
Flying squid	1	0.3-1	Surface at night	Lantern fish	Light and a jig	Jump	3
Skipjack	2	2-6	Surface	Squid, scad	Feather, spoon	Run	2.5
Deep king crab	2	2-8	Deep floor	Carrion	Deep pots	None	12
Gulper eel	2	1-3	Abyss	Shrimp	Deep-drop with a glow lure	None	10
Yellowfin tuna	3	15-80	Surface-mid	Squid, skipjack	Live skipjack, popper	Run, circle	5
Jumbo squid	3	10-40	Mid; rises at night in packs	Fish, squid	Glow jig	Jump, roll (inks)	3
Opah	3	20-80	Mid	Jellies, squid	Squid, fished deep	Circle	6
Frill shark	3	10-30	Deep	Squid	Deep-drop squid	Roll	5
Oarfish	3	20-100	Mid-deep	Plankton	Net only, rarely	None	300 flat per specimen
Swordfish	4	50-250	Deep by day, surface at night	Squid	Squid with a glow stick, at night	Dive, jump	7
Blue marlin	4	80-400	Surface	Tuna, squid	Trolled skirt lure, live skipjack	Jump, run	6
Giant bluefin	4	200-500	Surface-mid	Skipjack, squid	Live skipjack from the big-game chair	Run, circle	7
Atlantean coelacanth	4	40-90	Deep terraces	Fish, squid	Deep-drop dead bait	Dive	12

Species	Tier	Weight (kg)	Depth	Eats	Takes	Fight	Price (sh/kg)
Drowned-gold grouper	4	80–300	Terraces	Fish, crabs	Deep-drop live bait	Cover	9
Giant squid	4	100–300	Deep	Fish, squid	Deep-drop whole tuna	Dive, roll	8
Ocean sunfish	2	200–1,000	Surface	Jellies	Never bites; net bycatch	None	0
Sperm whale	4	—	Surface to abyss	Giant squid	Never	—	Never caught

Flora: sunken gardens on the terraces (shelter the grouper and coelacanth), deep coral (the crabs' habitat), the Eye's bioluminescent veils (drifting lights that rise through the night and draw lantern fish, and so everything above them).

Web notes. Atlantis has the longest chain in the game: lantern fish rise to the veils, squid rise to the lantern fish, tuna and swordfish rise to the squid, marlin hunt the tuna, and the Kraken takes whatever is largest. The sperm whale is a tell, not a catch: its clicks on the sonar mean a giant squid is below and the whale is hunting it, which briefly makes the deep safer for divers. Jumbo squid packs strip hooked fish and bite divers.

Threats

Threats are species in the food web with a size, a diet, senses, and a hunger meter, like everything else. What makes them threats is that the crew is on their menu. Each has a sonar tell, a sound tell, a visual tell, and at least one counter, all tested (see Definition of Done).

Threat	Grounds	Drawn by	Tells	What it does	Counters
Gull flock	Lagoon, Weeds	Fish on deck, chum	Cries overhead, shadows	Lands and flies off with fish under 3 kg	Shotgun, cover the catch, gut and ice fast
Barracuda pack	Lagoon	Flash lures, bleeding fish	Sonar: a tight fast group; silver flashes	Strikes hooked fish to the head (you reel in half); bites divers	Wire leader, land fish fast, spear the leader

Threat	Grounds	Drawn by	Tells	What it does	Counters
Reef or Great White shark	Lagoon (reef), Weeds, Atlantis	Blood above its threshold (30 reef, 120 White)	A fin at the lantern's edge; sonar: one slow large contact	Takes hooked fish; rams the hull; the White knocks crew off the rail	Harpoon cannon, full speed, stop the blood (ice the catch, stop chumming)
Siren	Weeds, Atlantis (the Deep Choir)	Light, singing back	Song rises; the helm drifts on its own	Charms the nearest crewman for 5 s (walks them to the rail); turns the helm toward rocks	Ear plugs, the bell breaks a charm, shoot the singer when it surfaces
Kelp Wraith	Weeds	Movement at the rail	Canopy moves against the current; nothing on sonar	Wraps a crewman at the rail and pulls them in	Stay off the rail in the canopy, cut them free with a knife, flare
Feral Mermen	Weeds	Nets in the kelp	Clicking on the hull mic	Cut nets and warps, steal from set gear	Haul early, rifle, trade them fish
Lantern Angler	Grotto, Atlantis	Light and curiosity	A second light in the dark that you didn't hang	Lures a crewman to the rail and pulls them over	Don't walk toward lights; hood the lantern; flare burst blinds it
Ghost Worm	Grotto, Atlantis	Vibration: taut lines, the net, the screw	Sonar: a long shape circling; lines hum	Bites through lines, snags the net, at full size coils the hull	Cut the line, stop the screw, lower the lantern; harpoon plus rifles on the head
Isopod swarm	Grotto	Corpses, offal on deck	Clicking on the anchor chain	Climb aboard, eat the catch, bite ankles	Raise the anchor, boiling water from the boiler bleed, clean deck
Drowned sailors	Grotto, Atlantis	Light, the bell of the Ghost Ship	Knocking on the hull, slow and regular	Climb aboard; slow, relentless, grab and drag	Gaffs and rifles, keep the deck lit, cast them back over
Cult longboat	Atlantis	The Pale Eye's Wake	Chanting from the dark, torches	Circles and chums the water around the <i>Gannet</i>	Outrun it, rifle the rowers, stop looking at the Eye

Threat	Grounds	Drawn by	Tells	What it does	Counters
Ghost Ship	Grotto (rare), Atlantis	Your bell rung too often; a moonless night	A bell answering yours	Comes alongside and boards with drowned crew; steals set gear	Flee at full steam, or fight: a beaten Ghost Ship leaves a relic chest
The Kraken	Atlantis	Wake at maximum	The Glass (dead calm), sonar whites out, the fish vanish	Arms take crew and gear off the deck, crush hull sections, can take the boat	Cut the net, kill the lantern, run at full steam, harpoon an arm to make it let go

Wake and the big three. The Great White, the Ghost Ship, and the Kraken are not rolled on a timer. They come when a ground's **Wake** (see The food web) crosses their threshold: 40 for the White, 65 for the Ghost Ship, 90 for the Kraken (Atlantis only). A careful crew can fish a whole night in Atlantis Waters without seeing the Kraken; a crew that depth-charges twice will see it before midnight.

Scaling by deadline. From the second deadline on, threats gain 10% hunger rate and 5% size per deadline, and each ground adds one threat from the next ground down the list, so the Lagoon eventually has Anglers.

Threat stats

Crew have no hit points (see Death), so a threat's effect on a crewman is an injury, a grab, or a trip over the rail. Hull damage is to one section's integrity (0-100). HP is the threat's; a weak point takes double damage. Values rise with the per-deadline scaling above.

Threat	Size	HP	Speed (m/s)	On the crew	On the boat and catch	Weak point	Drawn at
Gull flock	1 (×12)	1 each	8 in air	Pecks: no injury	Takes one fish under 3 kg every 4 s	Any hit	Fish on deck
Barracuda pack	3 (×4-8)	30 each	2, 7 burst	Bite on a diver (injury)	Bites hooked fish (-50% weight)	Head	Flash lures, blood 15
Reef shark	3	80	2, 5 burst	Bite at the waterline (injury)	Takes hooked fish	Gills	Blood 30

Threat	Size	HP	Speed (m/s)	On the crew	On the boat and catch	Weak point	Drawn at
Great White	5	600	3, 8 burst	Bite (serious injury); knocks crew off the rail	Ram: -25 to one section	Gills, eyes	Wake 40, blood 120
Siren	3	60	2	Charm for 5 s within 40 m of the song	Turns the helm 2° a second toward rocks	Head, when surfaced	Light, Wake 20
Kelp Wraith	3	50	1 (drifts)	Entangles; drags at 1 m/s toward the water	—	Body; a knife cut frees the crewman	Movement at the rail
Feral Merman	3	70	3	Thrown spear (injury)	Cuts a warp in 4 s; strips set gear	Head	Nets in the canopy
Lantern Angler	4	200	2, 8 lunge	Lures within 12 m (walks them to the rail); grabs	—	The lure: a hit blinds it for 20 s	Light, curiosity
Ghost Worm, hatchling	3	120	3	Bite (injury)	Bites through any line but steel	Head	Vibration
Ghost Worm, adult	6	1,500	3, 6 surge	Swallows a crewman in the water	Snags the net; coils the hull (-10 a second to two sections)	Three glowing gill slits	Vibration, Wake 55
Isopod swarm	1 (×30)	3 each	1	Ankle bites (slowed)	Each eats 1 kg of catch a minute	Boiler steam kills	Corpses, offal on deck
Drowned sailor	3	90	1.2	Grabs and drags at 0.8 m/s	—	Head; thrown overboard it climbs back in 30 s	Light, the Ghost Ship's bell
Cult longboat	Boat, 6 rowers	Boat 200, rowers 40 each	4	Thrown torches (burn)	Chums a ring of 20 blood a second	Rowers	The Pale Eye's Wake

Threat	Size	HP	Speed (m/s)	On the crew	On the boat and catch	Weak point	Drawn at
Ghost Ship	Ship	Hull 1,200	3.5	Boards with 6–10 Drowned sailors	Grapples alongside; steals set gear	Its bell	Wake 65, a moonless night
The Kraken	6	8 arms at 400, body 3,000	—	Arm grabs (carry off)	Crushes sections (–30), takes gear, can sink the boat	The eye, only when it surfaces	Wake 90, Atlantis

Most threats are worth something dead: a shark sells at 2 sh/kg, a Ghost Worm hatchling's gill lanterns fetch 60 each, an adult's 400. The Siren, Wraith, Drowned, and Kraken leave nothing sellable except relics on the Ghost Ship.

The big three

Each of the three Wake-called threats runs a short script on top of its food-web brain. Every one of them can be escaped, survived, or killed, and each takes a different kind of crew work. For reference, the *Gannet* makes 1.5 m/s at slow, 3 at half, and 4.5 at full (5.9 with the compound engine).

The Great White (Lagoon at deadline 2+, Weeds, Atlantis)

1. **Investigate (30–60 s).** A fin circles at the lantern's edge, about 20 m out. It takes any hooked fish it finds and bumps the hull once (–5). If blood near the boat falls below 60 and nobody is in the water, it leaves after 90 s.
2. **Hunt.** Starts if it is wounded, if a crewman is in the water, or if blood stays high. Every 20–30 s it drops out of sight (the sonar shows it rising, 2 s of warning) and rams: –25 to a section, 15° of roll, and unbraced crew at that rail go over. A crewman in the water is its first target, at 8 m/s.
3. **Break-off.** At 30% HP it flees, or it leaves once sated (60 kg eaten, a stolen tuna counts). Harpooned and tethered, it becomes a Run-pattern fight on the cannon winch; a landed White (300–700 kg) pays well and bleeds the whole ground.

The Ghost Ship (Grotto, Atlantis)

1. **The answer.** At Wake 65 on a moonless night, or after the ship's bell has been rung more than 10 times in one night, a bell answers from the dark. Fog thickens; the sonar shows a hull 300 m out.

2. **Approach (about 90 s).** It closes at 3.5 m/s, so full steam outruns it and half steam does not. Fleeing is a real option.
3. **Alongside.** It throws three grapples (each cut in 3 s with a knife) and boards Drowned sailors three at a time every 20 s, up to 6–10.
4. **Ending it.** Either a crewman crosses to the Ghost Ship and rings its own bell (the boarders fall into the sea), or the crew deals 1,200 hull damage (explosive harpoons and depth charges under it). Either way it sinks and leaves a floating relic chest: two relics and 300–600 in salvage.
5. **If ignored.** After 4 minutes alongside it casts off with any set gear and whatever un-iced catch its boarders carried.

The Kraken (Atlantis only)

1. **Tells (60 s).** The Glass: the wind dies and the sea goes flat. Fish vanish from the sonar. For the last 20 s the scope is white.
2. **Arms (0–60 s).** Two arms rise every 15 s, up to four. Each picks a target in order: the net, a crewman at the rail, the lantern mast, loose gear. A grabbed crewman is carried off in 4 s unless the arm takes 80 damage (knife 20 a hit, rifle 25, harpoon 120).
3. **Hold (60–90 s).** An arm on the warps drags the stern down; cutting the net loose buys 30 s. If the lantern is dark and the screw runs at full for 45 s straight, it loses the boat and sinks away.
4. **Crush (90 s on).** The remaining arms wrap the hull: –30 to a section every 10 s. Three sections at 0 sinks the *Gannet*. Every 30 s the body surfaces for 6 s and the eye (weak point) is open.
5. **Ending it.** Escape by darkness, speed, and cutting the net; survive by severing arms until it gives up (all eight arms cut); or kill it through the eye (3,000 HP, realistically only with explosive harpoons and depth charges). A kill leaves the Kraken's beak (a 2,000 relic) and ends every threat on the ground for the rest of the night.

Diving and salvage

Diving is The Trawl's Lethal Company facility run: one or two crew go down into a wreck for salvage while the deck keeps them alive. It is the highest-value, highest-trust activity in the game.

Gear. The hardhat suit (bought, 350) and later the diving bell (boat upgrade, 1,200). The hardhat diver is on an air hose from the deck pump and a lifeline to the gantry winch; the bell carries two divers to 120 m and gives them a pocket of air to return to.

The view. Diving switches that player to a side-view pixel scene built on the parkour side's movement code (walk, climb, squeeze, no jumping hazards), lit only by the helmet lamp. The deck crew sees the diver as a dot on sonar and hears them on the speaking tube.

Keeping the diver alive (the deck's job).

- **Air.** A hand works the pump (a rhythm input; the gauge must stay in the green). Stop pumping and the diver has 30 s of air in the helmet.
- **The hose and lifeline** are Verlet chains. They tangle on wreckage, get bitten (Ghost Worms, Barracuda), and snag if the boat drifts. The helm has to hold station over the wreck against wind and current.
- **Recall.** Two sharp tugs from the diver (a key) or the deck means "haul me up". The winch brings a diver up at 1 m/s; faster risks the Bends (slowed and dizzy for the rest of the night).

Wrecks. Each ground has 2–4 procedurally laid-out wrecks (smuggler hulls, whalers, Atlantean terraces) with rooms, flooded corridors, locked cabins (crowbar), and things living in them. Salvage items are physical: heavy, carried in both hands, sometimes two-diver lifts, sent up in the salvage basket on its own line.

Salvage. Brass fittings, bottles of old rum, ship's bells, chronometers, sealed strongboxes, scrimshaw, Atlantean idols, and **relics** (rare, worth 150–600 each). Some salvage is cursed: an idol brought aboard raises that night's Wake by 10 and makes the Pale Eye blink.

Wreck generation. Wrecks are generated per ground at the start of each deadline and persist across its three nights: what is looted stays looted, and what the crew disturbed stays awake.

Wreck type	Grounds	Depth	Rooms	Salvage budget	Locked cabins
Sloop	Lagoon, Weeds	8–20 m	3–5	80–200	0–1
Whaler	Weeds, Grotto	15–40 m	6–9	200–450	1
Smuggler hull	Grotto	20–60 m	8–12	300–700	2
Galleon	Atlantis	60–110 m (bell)	12–18	600–1,400	3
Atlantean terrace	Atlantis	40–120 m (bell)	10–16, open ruins	1–3 relics plus 300–800	1 sealed vault

Grounds hold Lagoon 1–2 wrecks, Weeds 2, Grotto 3–4, Atlantis 2–3 plus one terrace.

Layout rules

1. Start from a hull profile for the type and cut it into decks, then rooms on a 4×3 m cell grid joined by doors, hatches, and breaches. Some hulls lie on their side (30%), turning ladders into floors and floors into walls.
2. Every room must be reachable within the hardhat's 40 m of hose from the entry point, or from the bell for bell wrecks.
3. About 30% of wrecks have a squeeze passage (one diver, no carried salvage through it) that shortcuts to the richest room.
4. Place one air pocket per five rooms.
5. Spread the salvage budget with heavier and more valuable items deeper in; 40% of the value sits behind locked doors (crowbar or the Diver's Wreck Rat).
6. Silt on every floor: moving fast raises it and blinds the helmet lamp for 5 s.
7. Roll residents from the ground's roster: morays, octopus, isopods, congers, a Ghost Worm hatchling (Grotto, Atlantis), and one Drowned sailor per four rooms in the Grotto and Atlantis. Atlantis wrecks have a 10% chance per cabin of a cursed idol.

`--trawl-wreck <ground> <seed>` prints a wreck as ASCII and asserts reachability, the hose limit, and the salvage budget.

What lives in wrecks. Moray eels (bite and hold), octopus (steal the lamp), isopods, sleeping Ghost Worm hatchlings, and Drowned sailors who were only waiting for company.

Economy and progression

Money is shillings. Fish pay now and spoil; salvage keeps and pays best on deadline day. That split is the economy's main decision.

Fish value = species base price $\times$ weight $\times$ grade $\times$ freshness $\times$ glut.

- **Grade** is how it was taken (hook 100%, spear 85%, harpoon 75%, rifle 70%, depth charge 50%, explosive 30%), minus 10% for every bite taken out of it on the line.
- **Freshness** starts at 100% and falls 1% per real minute on deck, 0.2% per minute gutted and iced. Fish kept overnight in ice lose 25%; un-iced fish rot and sell for nothing.
- **Glut.** Each species' price at the Fish Market drops 3% for every 10 kg of it sold that deadline, so six hundred kilos of sardines is not a strategy.

Salvage is sold to the Owners at a rate that rises toward the deadline, the Lethal Company buy-back rule: 40% after night one, 70% after night two, 100% at the deadline count. Holding salvage is a bet that the crew will survive to sell it.

The Chandler. Consumables and hand gear, bought at the dock. Rods are priced in the Tackle table. The *Gannet* starts every run with two handlines, one light rod, two gaffs, a fish priest, one life ring, a patch kit, and 20 kg of ice.

Item	Price (sh)	Notes
Ice	5 per 20 kg	Keeps gutted fish fresh; the hold holds 600 kg
Coal	10 per sack	A sack runs the engine about 3 minutes at half
Squid strips, shrimp	4 per tin	10 baits
Chum bucket	15	Writes 40 blood over 60 s
Line spool (mono / braid / wire leader / glow)	8 / 15 / 12 / 20	One re-rig
Hooks (small, circle, big-game treble)	3 / 5 / 12	Per 5
Lures (spoon, jig, popper, plug, rattle, glow)	10–25 each	Lost with a broken line
Sabiki rig	12	Catches 3–6 bait fish at once
Longline	60	20 hooks, two radio buoys
Crab pot	25	Reusable
Hand lamp	20	Batteries last one night
Batteries	5	For lamps and walkies
Pocket watch	15	Shows the time anywhere
Walkie	40	Channel-wide voice
Knife	10	Cuts lines, grapples, arms, Wraiths
Crowbar	30	Opens locked cabins

Item	Price (sh)	Notes
Patch kit	25	Stops one leak
Extra life ring	40	Thrown on a rope
Ear plugs	8	Immune to Siren charm; halves hearing
Bandages	10 per 3	Stop bleeding
Flare pistol	60	Flares 10 each
Speargun	150	Spears 5 each
Rifle	250	Rounds 2 each
Shotgun	200	Shells 3 each
Harpoon (cannon)	20	Explosive head 80
Depth charge	120	Owners' fine 30 if used in the Lagoon
Hardhat diving suit	350	Hose and lifeline included

The Slipway (boat upgrades, kept for the whole run).

Upgrade	Effect	Price
Hull plates (per section)	Section integrity 100 → 150	150 each
Bigger ice hold	Hold 600 → 1,000 kg of iced fish	400
Second pump	Bilge rate doubled	300
Compound engine	+30% speed, -25% screw noise	800
Hooded lantern mast	Adds the searchlight level and colored filters (red light is invisible to most deep species)	500
Second sonar	A second operator; shows depth of contacts	450
Harpoon cannon	Bow cannon and winch	700
Big-game chair	The 120 kg rod station	900
Bigger trawl	Mouth width +50%, weight +50%	600
Diving bell	Two-diver bell to 120 m	1,200

The Catch Log. Every species caught is logged with its heaviest weight, and every threat with a one-line dossier learned from surviving it (the Lethal Company bestiary). A species' first catch pays a 50% bonus. The Catch Log, the heaviest-fish board, and trophies mounted in the wheelhouse carry across runs in the arcade profile.

Arcade tokens. 10 per deadline met, 25 for a run past the fourth deadline. Spent on cosmetics only (boat liveries, lantern colours, suit helmets), per the master reference: the Trawl never pays into the main game's economy.

Death, injury, and ghosts

Most deaths start with a crewman in the water, so the rescue is a mechanic, not a cutscene.

Overboard. A player in the water sees a side view of themselves treading water, has 25 s before cold and exhaustion drown them (12 s in a storm), and is a bleeding, splashing, noisy thing in the food web. The deck can throw the **life ring** (a thrown Verlet rope; aim for the swimmer, then haul), steer the boat alongside (the screw kills swimmers, so the helm must stop it), or send a second crewman in on a line. Things that took a crewman over the rail usually aren't done with them.

Injuries (no health bar). Bites, burns, and falls cause injuries that change what a player can do until the dock: a hooked hand (can't reel), a broken arm (one-handed only, can't carry big fish), a burn (slower), and a bite (bleeds into the scent grid until bandaged). A second serious injury in one night is death.

Death. A dead crewman is dead for the night. If their body stays aboard, they revive at the dock. If the body is lost to the sea, the Owners deduct 8% of the crew's money for the replacement hand (Lethal Company's rule), and the crewman still returns at the dock.

Ghosts. The dead walk the deck as translucent figures who cannot touch anything except three things: the ship's bell, the sonar screen (read only), and the ghost's own lantern, a faint cold light that only other ghosts can see. Ghosts see threats in the dark within 10 m as pale outlines for a moment after they move, so a dead crewman becomes the best lookout on the boat. They speak only to other ghosts and through the bell.

Total loss. If every crewman dies, or the boat sinks, the night's catch is lost, the Owners charge 25% of money for salvaging the *Gannet*, and the next night starts at the dock. The run continues unless the quota can no longer be met.

Technical plan

The Trawl is built on three existing pieces: the shared arcade networking layer, Red Tide's `Ecosystem` engine (extended from 3D to a 2.5D water column), and the parkour side's pixel-art and movement code.

Files

File	Holds
<code>trawl.cpp</code>	Session state machine (dock, sail, night, return, sell), quota, clock
<code>trawl_boat.cpp</code>	Hull, buoyancy, roll/pitch/heave, engine, damage, bilge
<code>trawl_lines.cpp</code>	Verlet chains for lines, nets, hoses, and ropes; tension, snapping, tangles
<code>trawl_fish.cpp</code>	Cast, bite, fight, landing; set gear; the trawl net
<code>trawl_guns.cpp</code>	Projectiles, refraction, damage, value grading
<code>trawl_dive.cpp</code>	Side-view dive scenes, wreck generation, air and hose
<code>trawl_eco.cpp</code>	Adapter to the shared <code>Ecosystem</code> : water-column depth bands, light layer, vibration, Wake
<code>trawl_art.cpp</code>	Procedural pixel art: deck, crew, fish, threats, water, lantern light
<code>trawl_data.cpp</code> + <code>trawl_species.json</code>	Every number and every species record
<code>trawl_bots.cpp</code>	Bot crew brains, station behaviours, orders

The water column. Red Tide's scent and sound grids are 3D at 2 m cells. The Trawl uses a 4 m horizontal grid with five depth bands (surface 0–2 m, upper 2–15, mid 15–60, deep 60–200, abyss 200+); agents keep a continuous depth for fishing and diving. Species records gain a depth band, a light preference, and a fight pattern.

Networking. Host-authoritative, 6 players, through `arcade_net.cpp`. Snapshots at 20 Hz; state is filtered by what each client can know: fish below the surface are sent only inside the light radius, on a client's sonar ping, or while hooked. The rod holder's tension gauge is

predicted locally from the chain physics and reconciled with the host so reeling feels immediate at 150 ms of latency. Voice uses the arcade layer's voice channel (Opus-encoded, positional), with the host mixing nothing: clients attenuate by distance and walls themselves. `--net-loop trawl` runs a host and five bot clients over loopback.

Performance budget. About 250 active agents within 150 m of the boat at full rate (10 Hz decisions, 60 Hz movement); population counters beyond. Up to 16 Verlet chains at once (6 rod lines, 2 warps, the net mesh counted as one, hose, lifeline, tether, life ring, anchor, longlines).

Art. Top-down pixel deck at 2× the parkour pixel scale; the sea is drawn black outside the light grid with foam and glint pixels; fish show as shadows until lit or landed, then as full sprites; threats get the parkour beast sprite pipeline. Side views (dive, overboard, the reel camera on a big fight) use the parkour renderer. The porthole bezel frames everything, per the arcade's shared frame.

Sound. Every cue goes through the master reference's cue registry and buses. Key cues: reel ratchet pitched by tension, the line's hum rising before a snap, the drag's zip, the winch, rod creak, the pump's rhythm and the diver's breathing, the bell, the hull mic's knocks, each threat's tell. The music is a drone and a shanty fragment that stops when anything with Wake-level size is within 60 m: the silence is the first tell.

Sound design

In the Trawl, sound is information: most threats are heard before they are seen, and the music going quiet is the first warning. Everything is synthesized through the master reference's cue registry and buses; where a Red Tide cue fits the same animal, it is copied as a starting point and re-tuned.

Music

State	Music
Dock	A concertina shanty with harbour bells; a new verse after each met deadline
Sail out and home	The shanty's refrain over the engine, slower at night's end
Quiet night	A low drone and a slow shanty fragment on a lone fiddle

State	Music
Fish on (anything over 20 kg)	A taut rising string figure that follows line tension
Wake-sized threat within 60 m	Silence. The music simply stops, which is the first tell
The big three engaged	A pounding low drum and hull-groan pulse; the Kraken adds a choir
Deadline met / missed	A brass fanfare on the chalkboard / a single church bell

Ambience by ground

Ground	Bed
Eclipse Lagoon	Surf on the crest, insects on the atoll, distant drums on canoe nights, gulls at dusk
The Weeds	Kelp creak, otters cracking urchins, sea lions far off, fog-horn buoys
The Grotto	Drips with long reverb, water sucking at the arch, mould hiss, the boat's every sound echoing back
Atlantis Waters	Open-ocean swell, wind, whale song far below, a low hum that rises when the Eye is watched

Effects

- **Fishing:** the reel's ratchet pitched by tension, the drag's zip, a rising line hum before a snap and the crack of the snap, the rod creaking, the splash of a jump, the thump of a gaff, the fish flopping on deck.
- **The boat:** engine chuff by telegraph setting, boiler hiss and the relief valve's shriek, hull creaks with roll, water in the bilge, the pumps, the winch, warps groaning under load, a snagged net's shriek.
- **Guns:** each weapon's report, muffled impacts under the surface, the harpoon's tether whine, the depth charge's thud felt through the hull.
- **Diving:** the pump's rhythm heard in the helmet, the diver's breathing (faster at low air), bubbles, hose scrape, silt, distant knocks inside the wreck.

Threat tells

Threat	Sound tell
Gulls	Cries overhead, closer each pass
Barracuda	Fast ticks on the hull mic
Sharks	Nothing, then a hull bump
Siren	Singing that rises through the music's silence
Kelp Wraith	Kelp creaking against the current
Mermen	Clicking on the hull
Lantern Angler	A low glassy hum at the second light
Ghost Worm	Lines humming with no fish on them
Isopods	Clicking on the anchor chain
Drowned sailors	Slow knocking on the hull, a few seconds apart
Cult longboat	Chanting and oars
Ghost Ship	A bell answering yours
Kraken	Everything goes quiet, even the sea

Voice

Proximity voice is processed in the world: muffled behind walls, thinned by wind and rain, lost under the engine at full, crackling on walkies, and bubbling for a crewman in the water. Ghosts hear each other with a cold reverb and hear the living faintly. Every sound tell has a visual cue for players who play without sound, and a subtitle toggle shows bot barks and telegraph messages as text.

Tests, balance, and build order

Self-tests on `depth.exe`

- `--trawl-sim <ground> <nights> [crew]` runs a bot crew of 1–6 (with a role per bot) and reports quota-met rate, money per night by source (hook, net, gun, dive), deaths by cause, and which threats arrived when.

- `--trawl-eco <ground> <minutes> [pattern]` runs the ecosystem alone with a scripted crew pattern (quiet, chumming, trawling, depth-charging) and prints population curves, blood, Wake, and threat arrival times.
- `--trawl-fight <species> <tackle> N` runs the fishing fight with a bot angler and reports land rate, mean fight time, and line breaks.
- `--trawl-shots` renders every species, threat, deck station, and weather to PNG.
- `--net-loop trawl` runs a host and five bots over loopback through a scripted night.
- `--trawl-wreck <ground> <seed>` prints a generated wreck and asserts its layout rules.

Balance targets (six-player bot crew, starting gear, first deadline)

Target	Value
Quota met, per ground	Lagoon 85%, Weeds 65%, Grotto 45%, Atlantis 25%
Deaths per night (mean)	Lagoon 0.3, Atlantis 2.5
Kraken seen per Atlantis night	Careful pattern under 10%, depth-charge pattern over 70%
Share of money from diving	20–35% by the third deadline
Medium rod vs 15 kg lingcod, bot angler	70–80% landed, 60–120 s fights
Mean session length	30–35 min

Build order

1. The boat alone: hull physics, engine, stations, top-down deck, movement, one player.
Gate: the boat rolls, lists under weight, and sinks when flooded.
2. Lines and fishing: Verlet lines, cast, bite, fight, landing against a dummy fish. Gate: `--trawl-fight` numbers in range.
3. Ecosystem port: the water column, light layer, vibration, Wake, the Lagoon's species.
Gate: `--trawl-eco lagoon` curves stable with no crew.
4. The session loop: dock, Chandler, sail, clock, sell, quota. Gate: a solo night on the Lagoon is playable end to end.

5. Networking: six players, voice, prediction for reeling. Gate: `--net-loop trawl` and a six-player LAN night.
6. Shooting, the net, set gear, death and ghosts.
7. The Weeds and the Grotto with their threats.
8. Diving and wrecks.
9. Atlantis Waters, the Kraken, the Ghost Ship.
10. Tuning pass and sound.

Definition of Done

- Six players over the internet complete a deadline including a dive, a Kraken encounter, an overboard rescue, and a death-and-ghost.
- Every threat has a distinct sprite, a sound tell, a visual tell, a sonar tell, and a counter proven by a scripted `--trawl-sim` case.
- The balance targets are met or deviations are logged in the balance changelog.
- Two players plus four bots complete a deadline on the Weeds, with bots working ordered stations and one bot taking over a dropped player's seat.
- Each role's perk is covered by a scripted `--trawl-sim` case (the Medic's CPR revive, the Angler's hook-set window, the Diver's air, the Bosun's patch time).

Designer decisions

Question	Decision	Where it lands
Voice	Built-in proximity voice is the main channel; walkies, the speaking tube, and the ping wheel supplement it	The crew of six: Talking
Grounds	The four drafted here (Eclipse Lagoon, the Weeds, the Grotto, Atlantis Waters) replace the master reference's five	The four grounds
Species and Red Tide	Engine code and some species may be shared; the Trawl's records, art, and systems are its own	The food web
Failed quota	The run ends on the first missed quota	The session loop

Question	Decision	Where it lands
Roles	Light roles with small perks: Bosun, Angler, Diver, Medic	The crew of six: Roles
Role upgrades	Three ranks per run, each a choice between two upgrades	The crew of six: Role upgrades
Crew size	1-6 players; the host can fill empty berths with bots	The crew of six: Bots
Master reference	Its Trawl section now points to this document, as it does for Fathoms	Depth master reference